

CE 211 – Quick Formula & Definitions Sheet

Key Cement Compounds

- C3S (Tricalcium silicate): early strength.
- C2S (Dicalcium silicate): later strength.
- C3A (Tricalcium aluminate): fast set, low sulfate resistance.
- C4AF (Tetracalcium aluminoferrite): minor effect.

Essential Reactions

- $C_3S + H \rightarrow C-S-H + Ca(OH)_2$ (fast, early strength).
- $C_2S + H \rightarrow C-S-H + Ca(OH)_2$ (slow, long-term).
- $Pozzolan + Ca(OH)_2 + H \rightarrow C-S-H$ (extra strength, durability).

Bogue's Equations (approx)

- $C_3S = 4.07CaO - 7.60SiO_2 - 6.72Al_2O_3 - 1.43Fe_2O_3 - 2.85SO_3$.
- $C_2S = 2.87SiO_2 - 0.75C_3S$.
- $C_3A = 2.65Al_2O_3 - 1.69Fe_2O_3$.
- $C_4AF = 3.04Fe_2O_3$.

Moisture Calculations

- Absorption (%) = $(W_{ssd} - W_{dry})/W_{dry} \times 100$.
- Total moisture (%) = $(W_{wet} - W_{dry})/W_{dry} \times 100$.
- Surface moisture (%) = Total – Absorption.
- SSD: pores full, surface dry → reference state for mix design.

Workability Tests

- Slump: measure settlement in mm (IS 1199).
- Vebe: time to remold under vibration (stiff mixes).
- Compacting factor = $W_t(\text{partial})/W_t(\text{full compaction})$.

Admixture Functions

- Plasticizer: +workability or –water (~5–15%).
- Superplasticizer: high-range water reduction (~25–35%).
- Accelerator: faster set/strength (e.g., $CaCl_2$; not for RC).
- Retarder: delays set (e.g., sugars, lignosulfonates).
- Air-entrainer: freeze-thaw resistance (3–6% air, strength ↓).

SCMs

- Fly ash: Class F (pozzolanic), Class C (pozzolanic + cementing).
- GGBS: low heat, durable, sulfate resistant.
- Silica fume: ultrafine, high strength/durability, needs SP.
- Rice husk ash: amorphous silica, variable quality.

Common Problems

- Segregation: separation of coarse/fines → fix grading, admixtures.
- Bleeding: water rise → reduce w/c, use air entrainment, good placement.